Raaghav Ramani

Contact Information

Center for Nonlinear Studies
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Employment

Los Alamos National Laboratory 2024–present

Mark Kac Postdoctoral Fellow in Applied Mathematics
Center for Nonlinear Studies

Education

University of California, Davis 2018–2023

Ph.D. in Applied Mathematics

Advisor: Prof. Steve Shkoller

Dissertation: *PDE-based methods for multiscale gas dynamics simulations*

Trinity College, University of Oxford 2012–2016

MMath Masters in Mathematics, Double First Class Honours

Advisor: Prof. Gregory Seregin

Thesis: *Regularity theory for the axially symmetric incompressible Navier-Stokes equations without swirl*

Research Interests

PDEs, fluid mechanics, computational physics.

Specifically: wave propagation, moving interface and free boundary problems, multiscale modeling, high-resolution WENO schemes, artificial viscosity, adaptive mesh generation, semi-Lagrangian and Arbitrary Lagrangian-Eulerian schemes, vortex methods, spectral methods, fluid instabilities, combustion, magnetohydrodynamics, computational shock formation and development, topological transitions.

Publications

R. Ramani
Interface tracking with Microscale Topological Surgery for two-dimensional filament breakup, submitted (2026)

R. Ramani
A boundary integral method for wave scattering in a heterogeneous medium with a moving obstacle, submitted (2026)

R. Ramani & S. Shkoller
A fast dynamic smooth adaptive meshing scheme with applications to compressible flow
Journal of Computational Physics, 490:112280 (2023)

R. Ramani & S. Shkoller
A multiscale model for Rayleigh-Taylor and Richtmyer-Meshkov instabilities
Journal of Computational Physics, 405:109177 (2020)

R. Ramani, J. Reisner, & S. Shkoller
A space-time smooth artificial viscosity method with wavelet noise indicator and shock collision scheme, Part 2: The 2-D case
Journal of Computational Physics, 387:45 – 80 (2019)

R. Ramani, J. Reisner, & S. Shkoller
A space-time smooth artificial viscosity method with wavelet noise indicator and shock collision scheme, Part 1: The 1-D case
Journal of Computational Physics, 387:81 – 116 (2019)

Fellowships & Awards

Mark Kac Postdoctoral Fellowship in Applied Mathematics 2024–2026
Los Alamos National Laboratory

Yueh-Jing Lin Scholarship 2023
Department of Mathematics, University of California, Davis

CSES Student Fellowship 2021–2023
Center for Space & Earth Science, Los Alamos National Laboratory

Graduate Research Fellowship 2019–2023
Department of Mathematics, University of California, Davis

Research Fellowship 2016–2018
Department of Mathematics, University of California, Davis

Met Office Masters Thesis Prize 2016
University of Oxford

Mitchell Scholarship 2014
University of Oxford

Scholarship 2013–2016
Trinity College, University of Oxford

Talks

Microscale Topological Surgery for tracking filament breakup
- CNLS Seminar, Los Alamos National Laboratory 06/2026

Adaptive front-capturing and front-tracking for multiscale fluid flows
- CNLS Seminar, Los Alamos National Laboratory 06/2025
- Multiphysics Models Conference, Los Alamos National Laboratory 06/2025

Computational Shock Development
- MultiMat Conference, Breckenridge 08/2024
- CASC Seminar, Lawrence Livermore National Laboratory 05/2024
- PDE & Applied Math Seminar, UC Davis 05/2024

PDE-based methods for multiscale computational fluid dynamics
- ANAG Seminar, Lawrence Berkeley Laboratory 09/2023
- Advanced Modeling and Simulation Seminar, NASA 09/2023
- Applied and Numerical PDEs Seminar, UC Berkeley 09/2023
- LANS Seminar, Argonne National Laboratory 08/2023

Efficient smooth adaptive meshing in multi-D with applications to gas dynamics
- Theoretical and Computational Astrophysics Network 05/2023
- T-3 seminar, Los Alamos National Laboratory 04/2023
- EES-16 seminar, Los Alamos National Laboratory 04/2023
- SOCAMS, Harvey Mudd College 05/2022

A PDE-based multiscale framework for efficient gas dynamics simulations
Student-Run Math and Applied Math Seminar, UC Davis 02/2023

Adaptive mesh methods for an ignition-based fire model
CSES symposium, Los Alamos National Laboratory 09/2022

A multiscale model for unstable interfaces in compressible flows with vorticity
Student-Run Math and Applied Math Seminar, UC Davis 05/2020

PDE methods for the numerical simulation of compressible fluid flow
Student-Run Math and Applied Math Seminar, UC Davis 04/2018

Conferences attended

SIAM Conference on Analysis of Partial Differential Equations (PD22) 03/2022
Virtual

Convex Integration and Nonlinear Partial Differential Equations 11/2021
ICMS, virtual

Advances and Challenges in Hyperbolic Conservation Laws 05/2021
ICERM, virtual

Mathematical Analysis of Geophysical Models and Data Assimilation 06/2020
Freie Universität Berlin, virtual

AMS Spring Central and Western Joint Sectional Meeting 03/2019
University of Hawaii at Manoa, Honolulu, HI, USA

Conference on Mathematical Analysis of Incompressible Fluids 06/2018
Mathematical Institute of University of Seville, Seville, Spain

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Teaching Experience

Spring	2023	MAT 119B	Ordinary Differential Equations
Winter	2023	MAT 128B	Numerical Analysis
Fall	2022	MAT 22B	Differential Equations
Spring	2022	MAT 21C	Calculus
Winter	2022	MAT 22B	Differential Equations
Fall	2021	MAT 22B	Differential Equations
Spring	2021	MAT 189	Advanced Problem Solving
Winter	2021	MAT 16B & 17A	Calculus (Lead TA)
Fall	2020	MAT 21A & 21C	Calculus (Lead TA)
Spring	2020	MAT 127C	Real Analysis
Winter	2020	MAT 127B	Real Analysis

Scientific Research & Other Work Experience

Graduate Student Researcher, Los Alamos National Laboratory 2022–2023
Group: Methods & Algorithms XCP-4
Adaptive mesh methods for an ignition-based fire model

Visiting scholar, University of California, Davis 2017–2018
Advisor: Prof. Steve Shkoller
PDE methods for computational physics

Undergraduate research student, University of Oxford 2015
Advisor: Prof. Ian Hewitt
Modelling of ice sheets

Summer school student, Perm State University 2015
Fluid Dynamics, Perturbation Theory, & Lie Group Analysis

	Undergraduate research student, University of Oxford Advisors: Prof. Steve Shkoller & Prof. Ian Hewitt *Computational methods for gas dynamics*	2014

Professional activities & service	Reviewer for Journal of Computational Physics	2022–present
	UC Davis student-run Analysis and PDE seminar, organizer	2020–2022
	American Mathematical Society, member	2018–present
	Society for Industrial and Applied Mathematics, member	2021–present
	Graduate Student Assembly, representative	2019–2020

Graduate Coursework

Real, complex, functional, harmonic analysis; linear & nonlinear PDEs; dynamical systems; calculus of variations; fluid mechanics; numerical methods for ODEs & PDEs; classical mechanics, quantum theory, relativity; algebra.

Relevant Skills

Computing:	Fortran, C++, MATLAB, Python, OpenMP, MPI, Kokkos
Languages:	English (fluent), Tamil (native)
Other:	First Aid, CPR, & AED certifications from American Heart Association

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